Dial In Number: 888-850-4523 Video: http://conf1.ls-cat.org
http://conf2.ls-cat.org

Passcode: 190580

9:00	Welcome / Approval of Minutes and Agenda (MG)
9:10	LS-CAT Status (KB)
	1. Rice joins 2. UIC would like to join (VOTE) 3. Bionanoprobe update 4. X-Ray microscopy workshop 5. Usage 6. APS Evaluation of LS-CAT Likely November 2013 7. APS Upgrade 8. Steps forward (and backward) 9. Reverse Remote Access 10. LS-CAT Budget Possibilities (DISCUSS and/or VOTE)
10:30 (?)	New Business / Next Meeting Planning

LS-CAT Status

Keith Brister

December 7, 2012

New Members

Rice University has joined LS-CAT as an Associate CAT Member in conjunction with the University of Wisconsin/Madison.

The University of Illinois at Chicago would like to join also, sharing UIUC's time. By voting to accept UIC as an Associate CAT Member at this time we obviate the need to resolve some UIUC/UIC/UI semantics in the LS-CAT Agreement.

VOTE NEEDED: Shall UIC be admitted to LS-CAT as an Associate CAT Member?

The Bionanoprobe at LS-CAT



August 2011: Offline commissioning

October 2011: Online commissioning, 50 nm features

November 2011: First frozen hydrated samples

February 2012: First tomography dataset

April 2012: First 24 nm images

October 2012: "Automated" tomography

APS and LS-CAT Personnel mostly involved:

Si Chen
Chris Roehrig
Stefan Vogt

Michael Bolbat
Jay VonOsinski
Keith Brister

X-Ray Microscopy Workshop

To promote use of the Bionanoprobe, Northwestern University, in collaboration with Argonne National Laboratory, The University of Chicago, and Xradia, will host a workshop August 9th and 10th at Northwestern's Chicago campus. Registration is free and attendance and encouraged for those labs that think they might be interested it using this instrument. It is likely that student travel grants will be available.

Visit the registration site at:

http://xrm2013.ls-cat.org

(There is a link on the LS-CAT webset ls-cat.org)

Recent Usage

Institution	Total Hours	Total Shots
University of Wisconsin-Madison	1354	174867
GSK/Van Andel Research Institute	1035	262550
Vanderbilt University	937	234087
Northwestern University	912	198255
University of Illinois at Urbana-Champaign	904	212108
University of Virginia	571	169091
MSU / Grand Valley State University	562	141587
University of Michigan	448	106442
Shamrock Structures	431	84428
Emerald / Wayne State University	368	111326
Nastrx, Inc	285	113741
Purdue University	195	44443
The University of Chicago	164	32441
The Scripps Research Institute-Florida	151	26997
University of Illinois at Chicago	129	17400
University of Toledo	70	11140
Southern Illinois University at Carbondale	67	10868
National Institutes of Health	63	23826
University of Cincinnati	54	10582
Universidad Nacional Autonoma de Mexico	43	31338
Rosalind Franklin University of Medicine and Science	42	10769
University of Massachusetts Medical School	37	13547
The Dow Chemical Company	32	7968
University of Southern California	20	5826
University of Toronto	19	10529
Baylor College of Medicine	17	3509
University of Oklahoma Health Sciences Center	11	997
Marquette University	10	3103
University of Ottawa	8	1930
Xtal BioStructures Inc.	7	1439
Food and Drug Adminstration	2	114

Numbers are for the runs 2011-3, 2012-1, and 2012-2: October 2011 through August 2012.

Allocation for this time period was 1661 hours/share

Statistics for crystallography only. BNP uses the GU time on 21-ID-D

GU usage by hours is 27% of the total.

Disclaimers:

Usage can exceed allocation by requesting rapid access slots one week ahead of time.

Times are based on first and last image times and do not include setup/teardown or thinking about what you are doing.

APS Evaluation of LS-CAT

We have been informed to expect that the APS Scientific Advisory Committee (SAC) will review LS-CAT in November 2013. While we do not expect any problems, it is very much to our advantage to make a very good impression and to treat this as seriously as a major funding review. The exact date has not been set.

Review will include (among other things):

- Complete publication list for all members and GU's
- Complete PDB deposit list for all members and GU's
- Presentations by several members and probably 1 GU
- Plans for the future new science
- Review of funding and staffing

This will be the most rigorous scientific review we will receive before starting negotiations on extending the LS-CAT agreement.

APS Upgrade

Yesterday the APS finished its CD-2 Lehman review.

- This is an important step in the it sets the baseline cost and schedule for the project.
- It will likely be several months before the DOE gives its approval.

In preparation for the continuous 150 mA running planned for in the upgrade, the APS conducted a test in August that allowed certain beamlines known to be engineered to accept high heat loads to accept beam while the storage ring was ramped up to 150 mA. LS-CAT, a suitably modern beamline, participated.

We found no deleterious effects on any of our beamlines. We are ready now for high current running.

New XRAY Beam Position Monitors have been designed to stabilize the undulator beams. The cost is high ($500K by current estimates) but may be worth it.

Steps Forward (1)

Our MD-2 control computers and software are becoming obsolete. They rely on Windows XP Pro, a Matrox PCI frame grabber card, a motherboard that supports said frame grabber card, an analog camera, and .NET 2003 Visual Studio. This means that we are vulnerable to hardware failures and cannot easily add new features. Here are the options:

1. Purchase new MD-2 control software from Maatel and re-engineer our MD-2 control interface.

2. Develop a replacement control program and use a version of our remote access program to control the MD-2 locally

We are not the only ones in this situation as both NE-CAT and SER-CAT have similar hardware.

Steps Forward (2)

Recent workshops sponsored by the APS Life Sciences Council brought together scientists from each of the biological beamlines to talk about, among other things, approaches to controlling MD2 diffractometers and supporting remote access users.

Using approaches demonstrated by GM/CA and SER-CAT I have developed an MD-2 control program that allows us to collect data much faster: 23 frames per minute for 1 second exposures instead of our current 18 frames per minutes. This, together with a data sharing approach used by NE-CAT, will allow us to implement an improved remote access client during 2013 that includes greater data collection efficiency and some much requested features.

Currently we are working with NE-CAT to roll out this new MD-2 control program and to bring their RAPD data analysis package to LS-CAT.

Steps Forward (3)

Tentative rollout schedule

1. January down period: Install new control infrastructure in parallel with our existing infrastructure.

2. 2013-1: test on our live stations collecting real data with real users. This setup allows us revert at any time back to the current MD-2 computers.

3. 2013-1 (also): Test new cameras on 21-ID-E. This precludes reverting back to the old MD-2 controls on this station.

4. May down period: replace cameras on MD-2 computers.

5. 2013-2: run on new hardware and software.

Steps Backward

New low level robot software was purchased from IRELEC, our sample mounter robot vendor. This new software was supposed to speed up things by "10-15%" along with a few other improvements. It does this by keeping the robot moving as it passes through a number predefined points in a given path as opposed to pausing at each one.

Unfortunately the software delivered proved buggy and disastrous for operations and had to be "reverted" after a week or so at the beginning of the Fall 2012 run.

At this point we'll work with the vendor during the upcoming down period to debug the installation. Without optimization 10%–15% might be optimistic though we should be able to realize some gains.

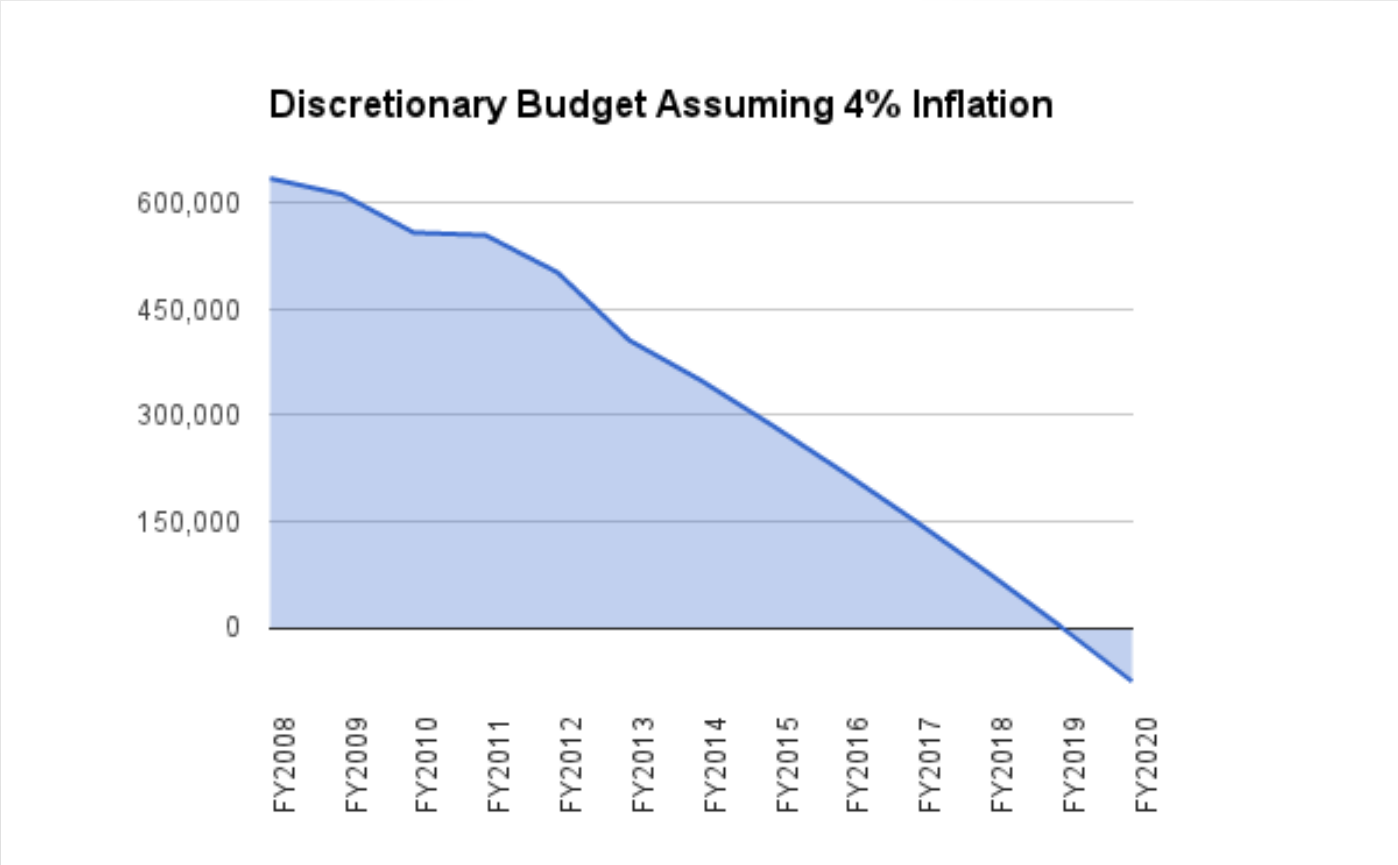
Treading Water

It's time to replace the compressors on our Rayonix (MARCCD) detectors on F and G. We've just done this for D and are now starting to see failures on F and G. We have been fortunate in that we've not seen any significant down time due to this. (around $70K to $100K)

Budget Forecasts

Budget Forecasts

Assuming dues remain at current levels ($300K/Share) then at some point we'll need to start burning our saved capital to support operations. Likely this will be by FY 2015, probably in FY 2014.



Budget Forecasts

Assuming the same number of LS-CAT shares each paying $300K/year, no new money, no new detectors, not commissioning 21-ID-E, no major equipment failures, no changes in the overhead or fringe rates, and no personnel changes we will run out of money between 2019 and 2022:

Assumed Inflation Rate	Year
2.5%	2022
3.0%	2021
4.0%	2020
5.0%	2019

Inflation Adjusted Dues

What to do? On going operations are going to require more funds than we are currently billing. I recommend future budget targets at least allow for inflation. Here is a schedule based on 4% inflation:

Fiscal Year	Budget	Dues/share
(current)	1,950,000	300,000
2014	2,028,000	312,000
2015	2,109,120	324,480
2016	2,193,485	337,459
2017	2,281,224	350,958
2018	2,372,473	364,996
2019	2,467,372	379,596
2020	2,566,067	394,780
2021	2,668,710	410,571
2022	2,775,458	426,994

Two possible FY14 budgets: Flat and a 4% increase.

Description	FY2013 Budget	FY14 FLAT	FLAT-FY13 Bdgt Diff	FY14 4%	4% Bdgt - FLAT
Academic Personnel	417,298	445,717	28,419	445,851	
Non-Academic Personnel	120,285	125,253	4,968	125,253	
Fringe Benefits	228,488	241,588	13,100	241,588	
Salaries Restricted	309,841	324,521	14,680	324,521	
Supplies	76,000	62,000	-14,000	67,000	5,000
Supplies, Restricted	5,000	4,100	-900	4,100	
Services	66,000	13,000	-53,000	13,000	
Services, Restricted	12,000	20,000	8,000	20,000	
External Space Rental	109,935	114,332	4,397	114,332	
Travel, Domestic	2,000	5,000	3,000	25,000	20,000
Utilities & Telephone	8,000	6,000	-2,000	6,000	
Capital Equipment	208,000	200,000	-8,000	245,000	45,000
Indirect Cost - Sponsored	387,153	388,489	1,335	396,356	7,867
Totals:	1,950,000	1,950,000		2,028,000	78,000
Per Share	300,000	300,000		312,000	12,000

Push for reverse remote access and a presence at national and international meetings to increase value to members and attract new interest in LS-CAT.

Increased in money available for maintenance and small scale innovation.

Capital Projects

Future needs to keep LS-CAT crystallography relevant:

2 PAD detectors	4,000,000
E Mono	500,000
New Focusing System for D	500,000
XGRID BPM	500,000
Optimized Undulators	2,000,000
Total:	**$7,500,000**

Other Funding

Possibilities for federal funding:

NIH declined to fund our proposal for new fast detector. We can, and should, try again. Just saying we want to do the same science only faster is not enough to obtain funding, we need to put forth new science that relies on new detector technology.

An NSF proposal is not out of the question. Judging by the list of successful NSF proposals it's unlikely to be successful if it only covers crystallography uses.

We are not the only CAT that has unfunded needs. I am currently working with the other sectors to see what an APS wide biological proposal to the major funding agencies would look like. Probably a long shot but the idea has a some traction.